



Semester II 2022/2023

Subject : Bioinformatics I (SECB2103)
Section : 01 – Dr Haslina Hashim
Topic : Lab 02 – Access to Biological Databases

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1. The purpose of this problem is to introduce you to using Entrez and related NCBI resources.

a) How many human proteins are bigger than 300,000 daltons?

5214 human proteins are bigger than 300,000 daltons.

b) What is the longest human protein?

The longest human protein is Titin.

There are several different ways to solve these questions.

(1) Try to first limit your search to human by using TaxBrowser. From the home page of NCBI select the alphabetical list of resources and find the Taxonomy Browser and the entry for human. Then follow the link to Entrez Protein, where all the results will be limited to human.

(2) Enter a command in the format xxxxxx:yyyyyy[molwt] to restrict the output to a certain number of Daltons; for example, 002000:010000[molwt] will select proteins of molecular weight 2,000 to 10,000.

(3) As a different approach, search 30000:50000[Sequence Length]

(4) You can read more about titin (NP_596869.4), the longest human protein, in an NCBI newsletter (WebLink 2.73 <ftp://ftp.ncbi.nih.gov/genbank/gbrel.txt>). While the average protein has a length of several hundred amino acids, incredibly titin is 34,423 amino acids in length.

(5) Explore additional ways to limit Entrez searches by using an NCBI Handbook chapter (<http://www.ncbi.nlm.nih.gov/books/NBK44864/>)

Write a report of how you would answer the question in this week's session.

Include screen captures as well as detailed explanation and observations.

All screen captures should be labelled as figures and numbered accordingly.

All tables should be labelled as tables and numbered accordingly.

Hint: What is the Latin name for humans? Homo sapiens.

1. Answer :

First, open taxonomy browser and search for human.

The screenshot shows the NCBI Taxonomy Browser interface. The search bar contains 'human' and the results are displayed as a list of links to various taxonomic levels. The results are organized into a table with three columns: Archaea, Bacteria, and Eukaryota. The table lists various organisms, including Arabidopsis thaliana, Bos taurus, Caenorhabditis elegans, Chlamydomonas reinhardtii, Danio rerio (zebrafish), Dictyostelium discoideum, Drosophila melanogaster, Escherichia coli, Hepatitis C virus, Homo sapiens, Mus musculus, Mycoplasma pneumoniae, Oryza sativa, Plasmodium falciparum, Pneumocystis carinii, Rattus norvegicus, Saccharomyces cerevisiae, Schizosaccharomyces pombe, Takifugu rubripes, Xenopus laevis, and Zea mays. The interface also includes a disclaimer, a reference, and a comment section.

NCBI Taxonomy Browser

Search for [human] as | complete name | lock Go Clear

Display 3 levels using filter: none

The "Token set" option returns longer names that include the search terms, e.g., hybrid taxa. See what happens if you query "Bos taurus" using the "Complete match" option versus the "Set of tokens" option. The "Phonetic search" option can be used when you are not sure about the exact spelling of a organism name. It tries to find the phonetically closest strings (try "Drozofila" as an example).

This is the top level of the taxonomy database maintained by NCBI/GenBank. You can explore any of the taxa listed below by clicking it.

- Archaea
- Bacteria
- Eukaryota
- Viruses
- Other
- Unclassified

These are direct links to some of the organisms commonly used in molecular research projects:

Arabidopsis thaliana	Escherichia coli	Pneumocystis carinii
Bos taurus	Hepatitis C virus	Rattus norvegicus
Caenorhabditis elegans	Homo sapiens	Saccharomyces cerevisiae
Chlamydomonas reinhardtii	Mus musculus	Schizosaccharomyces pombe
Danio rerio (zebrafish)	Mycoplasma pneumoniae	Takifugu rubripes
Dictyostelium discoideum	Oryza sativa	Xenopus laevis
Drosophila melanogaster	Plasmodium falciparum	Zea mays

Disclaimer: The NCBI taxonomy database is not an authoritative source for nomenclature or classification - please consult the relevant scientific literature for the most reliable information.

Reference: How to cite this resource - Schoch CL, et al. NCBI Taxonomy: a comprehensive update on curation, resources and tools. Database (Oxford). 2020: [baaa062](#). PubMed: [32761142](#) PMC: [PMC7408187](#).

Comments and questions to info@ncbi.nlm.nih.gov

Then, click on homo sapiens.

The screenshot shows the NCBI Taxonomy Browser interface. The search bar at the top contains 'Homo sapiens'. Below the search bar, the taxonomic details for 'Homo sapiens' are displayed, including the Taxonomy ID (9606), current name, and various links to related databases. On the right side, there is a table titled 'Entrez records' which lists various database entries and their corresponding counts.

Database name	Subtree links	Direct links
Nucleotide	28,919,848	28,918,446
Protein	1,997,862	1,997,303
Structure	71,787	71,783
SNP	1,191,662,003	1,191,662,003
Conserved Domains	82	82
GEO Datasets	4,204,281	4,204,281
PubMed Central	63,397	63,314
Gene	260,077	260,004
HomoloGene	18,713	18,713
SRA Experiments	5,691,042	5,689,891
GEO Profiles	61,958,910	61,958,910
Protein Clusters	13	13
Identical Protein Groups	2,281,364	2,281,271
BioProject	125,548	125,509
BioSample	11,577,633	11,577,370
Datasets Genome	2,129	2,129
dbVar	8,129,185	8,127,852
Genetic Testing Registry	68,275	68,275
PubChem BioAssay	847,316	847,309
Taxonomy	2	1

Next, on the above page, click the protein link. This allows to link to proteins that are human.
The taxonomy identifier (txid)9609 corresponds to human.

The screenshot shows the NIH National Library of Medicine Protein database interface. The search bar at the top contains 'txid9606[Organism:exp]'. Below the search bar, the taxonomic details for 'Homo sapiens' are displayed, including the Taxonomy ID (9606) and a list of related proteins. The page also includes a sidebar with various filters and a table of related proteins.

Species	Summary	Sort by	Send to	Filters
Animals (1,997,766)	Summary	20 per page	Sort by Default order	Manage Filters
Fungi (19)				
Bacteria (3)				
Viruses (2)				
Customize ...				
Source databases				
PDB (216,323)				
RefSeq (199,081)				
UniProtKB / Swiss-Prot (20,596)				
Customize ...				
Genetic compartments				
Mitochondrion (831,074)				
Plasmid (78)				
Plastid (3)				
Sequence length				
Custom range...				
Molecular weight				
Custom range...				
Release date				
Custom range...				
Revision date				
Custom range...				

On the left sidebar select the molecular weight range from 300,000 to 600,000.

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

Protein

Create alert Advanced Help

Species: Animals (5,741) Customize ...

Source databases: PDB (447) RefSeq (3,265) UniProtKB / Swiss-Prot (210) Customize ...

Sequence length: Custom range...

Molecular weight: ☒ From 300000 to 600000 (5,741)

Release date: Custom range...

Revision date: Custom range...

Clear all Show additional filters

TAXONOMY

Was this helpful?

Homo sapiens

Human (*Homo sapiens*) is a species of primate in the family *Hominidae* (great apes).
Taxonomy ID: 9606

Filter your results: All (5741) [Manage Filters](#)

Find related data
Database:

Search details
txid9606[Organism:noxp] AND ((000300000[MOLWT] : 006000000[MOLWT]))
 [See more...](#)

Recent activity
[Turn Off](#) [Clear](#)

txid9606[Organism:noxp] AND ((300000[MOLWT] : 600000[MOLWT]) Protein
txid9606[Organism:noxp] AND ((300000[MOLWT] : 600000[MOLWT]) Protein

Items: 1 to 20 of 5741

<< First < Prev Page 1 of 288 Next > Last >>

Filters activated: Molecular weight from 300000 to 600000. [Clear all](#)

☐ [serine/threonine-protein kinase ATR isoform 1 \[Homo sapiens\]](#)

1. 2644 aa protein
Accession: NP_001175.2 GI: 157266317
[BioProject](#) [Nucleotide](#) [PubMed](#) [Taxonomy](#)
[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

This show the result, there are 5741 proteins > 3000,000 daltons.

Customize ...

Sequence length: Custom range...

Molecular weight: ☒ From 3900000 to 6000000

Release date: Custom range...

Revision date: Custom range...

Clear all Show additional filters

TAXONOMY

Was this helpful?

Custom range: to

ily *Hominidae* (great apes).

Items: 5

Filters activated: Molecular weight from 3900000 to 6000000. [Clear all](#)

☐ [titin isoform IC \[Homo sapiens\]](#)

1. 35991 aa protein
Accession: NP_001254479.2 GI: 642945631
[BioProject](#) [Nucleotide](#) [PubMed](#) [Taxonomy](#)
[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

☐ [titin isoform X1 \[Homo sapiens\]](#)

2. 35622 aa protein
Accession: XP_054199629.1 GI: 2462576605
[BioProject](#) [Nucleotide](#) [Taxonomy](#)
[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

☐ [titin isoform X1 \[Homo sapiens\]](#)

3. 35622 aa protein
Accession: XP_016860308.1 GI: 1034616263
[BioProject](#) [Nucleotide](#) [Taxonomy](#)
[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

☐ [titin \[Homo sapiens\]](#)

4. 35991 aa protein
Accession: KAI4037134.1 GI: 2247825226
[BioProject](#) [Nucleotide](#) [Taxonomy](#)

Analyze these sequences
[Run BLAST](#)
[Align sequences with COBALT](#)
[Identify Conserved Domains with CD-Search](#)

Find related data
Database:

Search details
txid9606[Organism:noxp] AND ((003900000[MOLWT] : 006000000[MOLWT]))
 [See more...](#)

Recent activity
[Turn Off](#) [Clear](#)

txid9606[Organism:noxp] AND ((3900000[MOLWT] : 6000000[MOLWT]) Protein
txid9606[Organism:noxp] (1997303) Protein
Pan troglodytes AND (chimpanzee) AND (aive[prop]) (42266) Gene
chimpanzee globin (0) Taxonomy
HBB hemoglobin subunit beta [Pan troglodytes] Gene

See more...

Here shows titan is the longest human protein.

2. The purpose of this problem is to obtain information from the NCBI website.

The RefSeq accession number of human beta globin protein is NP_000509. Go to NCBI

(<http://www.ncbi.nlm.nih.gov/>).

a) What is the RefSeq accession number of beta globin protein from the chimpanzee (*Pan troglodytes*)?

XP_508242.1

(1) There are several different ways to solve this. Try typing chimpanzee globin into the home page of NCBI; or use the Taxonomy Browser to find chimpanzee Entrez Gene entries.

(2) HomoloGene (<http://www.ncbi.nlm.nih.gov/homologene>) (WebLink 2.38) is a great resource to learn about sets of related eukaryotic proteins. Use HomoloGene to find a set of beta globins including chimpanzee.

2. Answer

Entering chimpanzee globin into the NCBI home page leads to results in Gene

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

Search NCBI **Search**

Results found in 12 databases

Literature	Genes	Proteins
Bookshelf 17	Gene 28	Conserved Domains 0
MeSH 0	GEO DataSets 1	Identical Protein Groups 14
NLM Catalog 0	GEO Profiles 1	Protein 108
PubMed 122	PopSet 0	Protein Family Models 0
PubMed Central 1,191		Structure 0

Genomes	Clinical	PubChem
Assembly / Genome NCBI Datasets	ClinicalTrials.gov 0	BioAssays 0
BioCollections 0	ClinVar 0	Compounds 0
BioProject 0	dbGaP 0	Pathways 0
BioSample 0	dbSNP 0	Substances 0

2°C partly sunny

The Gene result sorts chimpanzee HBB to the top:

NIH National Library of Medicine
National Center for Biotechnology Information

Gene **Search**

Create RSS Save search Advanced Help

Gene sources Genomic

Categories Alternatively spliced Annotated genes Protein-coding

Sequence content Ensembl RefSeq

Status ☒ Current

Clear all Show additional filters

Tabular 20 per page Sort by Relevance Send to

See [Globin globin-2B](#) in the Gene database
globin reference sequences [Transcript \(4\)](#) [Protein \(4\)](#)

Search results
Items: 1 to 20 of 28
See also 4 discontinued or replaced items.

Name/Gene ID	Description	Location	Aliases
☐ HBB ID: 450978	hemoglobin subunit beta [<i>Pan troglodytes</i> (chimpanzee)]	Chromosome 9, NC_072407.2 (9357035..9358653, complement)	CK820_G0005984, GLNA1
☐ HBD ID: 748577	hemoglobin subunit delta [<i>Pan troglodytes</i> (chimpanzee)]	Chromosome 9, NC_072407.2 (9364385..9366163, complement)	CK820_G0005983, GLNA2
☐ HBG2 ID: 450979	hemoglobin, gamma G [<i>Pan troglodytes</i> (chimpanzee)]	Chromosome 9, NC_072407.2 (9384837..9386297, complement)	CK820_G0005946
☐ HBG1 ID: 736917	hemoglobin, gamma G [<i>Pan troglodytes</i> (chimpanzee)]	Chromosome 9, NC_072407.2 (9379911..9381362, complement)	CK820_G0005980, HBG2
☐ HBA1 ID: 732485	hemoglobin subunit alpha 1 [<i>Pan troglodytes</i> (chimpanzee)]	Chromosome 18, NC_072416.2 (2508555..2509388)	CK820_G0054686, HBA2
☐ HBE1 ID: 100190972	hemoglobin subunit epsilon 1 [<i>Pan troglodytes</i> (chimpanzee)]	Chromosome 9, NC_072407.2 (9399914..9654572, complement)	CK820_G0005962, GLNB2
☐ HBA2	hemoglobin subunit alpha 2 [<i>Pan troglodytes</i> (chimpanzee)]	Chromosome 18, NC_072416.2	CK820_G0054685

Filters: [Manage Filters](#)

Results by taxon

Find related data Database: Select

Find items

Search details
[("Pan troglodytes"[Organism] OR chimpanzee[All Fields]) AND globin[All Fields]] AND alive[prop]

Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share

12°C partly sunny

Follow the first entry (HBB) to see the RefSeq accession. In the above screen shot you can also see the accession NC_072407.2. This corresponds to the entire chromosome 9 of the chimp.

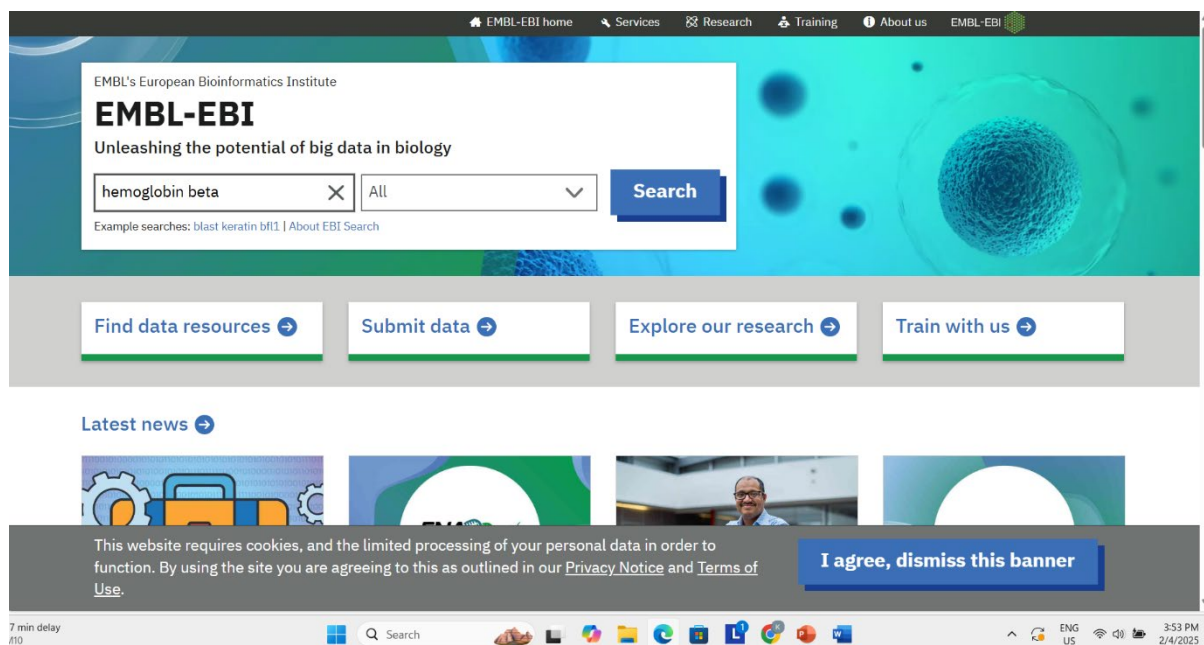
3. The purpose of this exercise is to become familiar with the EBI website and how to use it to access information.

(1) Visit the site (<http://www.ebi.ac.uk/>) (WebLink 2.5). Enter hemoglobin beta in the main query box (alternatively use the query human hemoglobin beta).

(2) Inspect the results. Explore the various links to information about pathways, genomes, nucleotide and protein sequences, structures, protein families, and more.

3. Answer

Enter hemoglobin beta at the EBI home page:



Here is the search result page:

EMBL's European Bioinformatics Institute

EBI Search

Access all EMBL-EBI resources

hemoglobin beta

Search

Examples: VAV_HUMAN, tp53, Sulston...

Advanced search

Search results for **hemoglobin beta**

Showing **51** results out of **44,674** in All results

[Give us feedback on these results](#)

Filter your results

Source

All results (44,674)

[Genomes & metagenomes \(1,397\)](#)
[Nucleotide sequences \(9,307\)](#)

Protein sequences (9,726 results)

Source: UniProtKB (ID: HBB_HUMAN)

P68871

Hemoglobin subunit beta

Homo sapiens(Reviewed)

Secondary accession number(s): A4GX73, B2ZUE0, P02023, Q13852, Q14481, Q14510, Q45KT0, Q549N7, Q6F108, Q6R7N2, Q8IZI1, Q9BX96, Q9UCD6, Q9UCP8, Q9UCP9

Here can explore the various links to information about pathways, genomes, nucleotide and protein sequences, structures, protein families, and more:

Filter your results

Source

All results (44,674)

[Genomes & metagenomes \(1,397\)](#)
[Nucleotide sequences \(9,307\)](#)
[Protein sequences \(9,726\)](#)
[Macromolecular structures \(2,015\)](#)
[Bioactive molecules \(23\)](#)
[Gene expression \(491\)](#)
[Diseases \(134\)](#)
[Molecular interactions \(199\)](#)
[Gene-Disease Associations \(2,711\)](#)
[Reactions & pathways \(35\)](#)
[Protein families \(51\)](#)
[Protein expression data \(234\)](#)
[Enzymes \(9\)](#)
[Literature \(18,122\)](#)
[Samples & ontologies \(173\)](#)
[EMBL-EBI website \(5\)](#)
[Other data \(42\)](#)

Protein sequences (9,726 results)

Source: UniProtKB (ID: HBB_HUMAN)

P68871

Hemoglobin subunit beta

Homo sapiens(Reviewed)

Secondary accession number(s): A4GX73, B2ZUE0, P02023, Q13852, Q14481, Q14510, Q45KT0, Q549N7, Q6F108, Q6R7N2, Q8IZI1, Q9BX96, Q9UCD6, Q9UCP8, Q9UCP9

Cross references: [Protein expression data \(1,019\)](#) [Macromolecular structures \(293\)](#) [Nucleotide sequences \(208\)](#) [+ show more](#)

Formats: [in FASTA format](#) [in Feature Viewer](#) [in Interpro Matches](#) [+ show more](#)

Source: UniProtKB (ID: HBA_HUMAN)

P69905

Hemoglobin subunit alpha

Homo sapiens(Reviewed)

Secondary accession number(s): P01922, Q1HDT5, Q3MIF5, Q53F97, Q96KF1, Q9NYR7, Q9UCM0

Cross references: [Protein expression data \(964\)](#) [Macromolecular structures \(299\)](#) [Protein sequences \(147\)](#) [+ show more](#)

Formats: [in FASTA format](#) [in Feature Viewer](#) [in Interpro Matches](#) [+ show more](#)

Source: UniProtKB (ID: HBD_HUMAN)

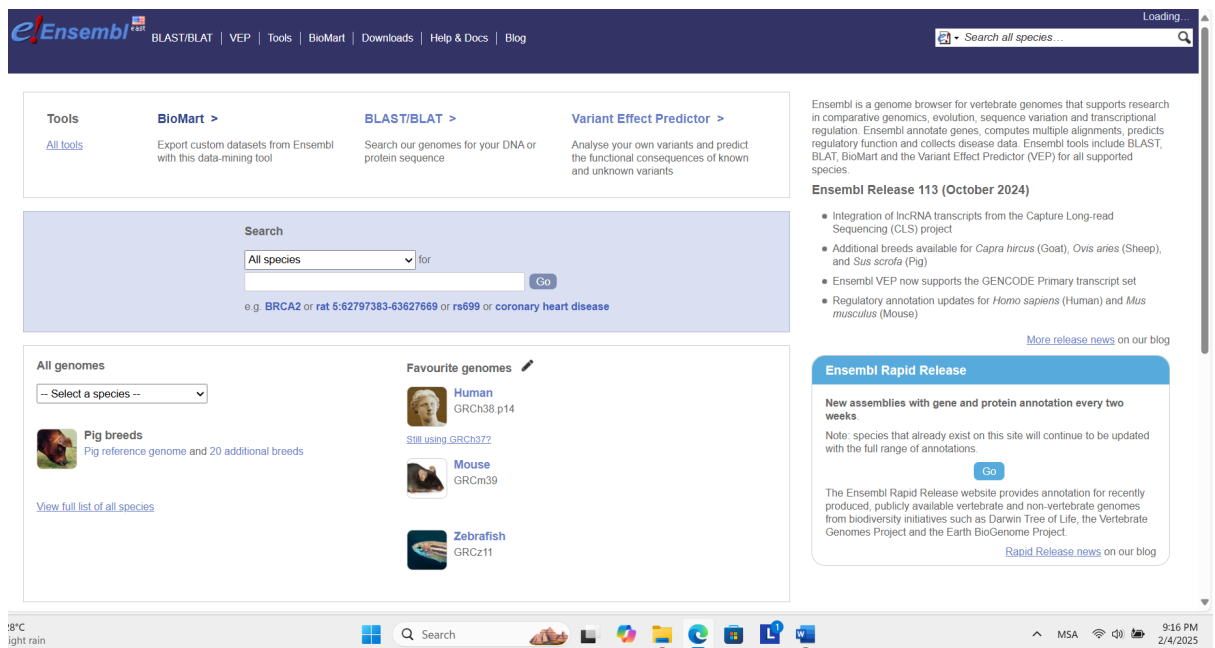
P02042

4. Accessing information from BioMart: the beta globin locus

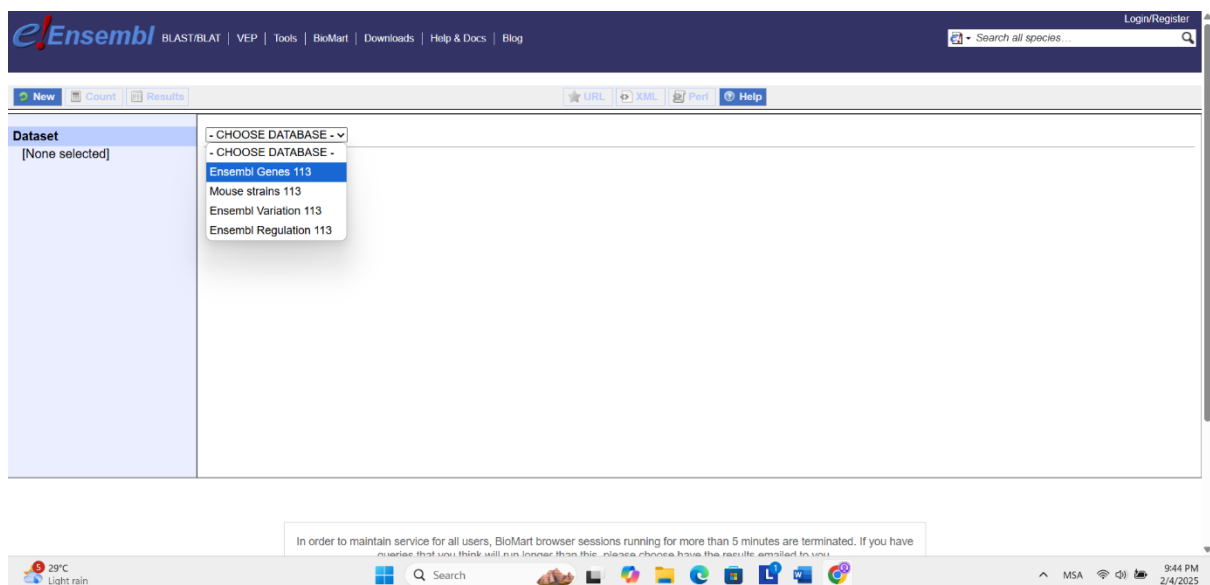
- (1) Go to www.ensembl.org and follow the link to BioMart.
- (2) First choose a database; we will select Ensembl Genes 71.
- (3) Choose a dataset: Homo sapiens genes (GRCh37.p10). Note the other available datasets.
- (4) Choose a filter. Here the options include region, gene, transcript event, expression, multispecies comparisons, protein domains, and variation. Select “region”, chromosome 11, and enter 5240000 for the Gene Start (bp) and 5300000 (bp) for the Gene End. (Note that this region spans 60 kilobases and corresponds to chr11:5,240,001-5,300,000.)
- (5) Choose attributes. Select the following features. Under “Gene” select Ensembl Gene ID and %GC content; under “External” select the external references CCDS ID, HGNC symbol (this is the official gene symbol) and HGNC ID(s).
- (6) At the top left select “Count.” Currently there are 8 genes matching these criteria.
- (7) To view these results press “Results.” Note that you can export your results in several formats (including a comma separated values or CSV file) that can be further manipulated (e.g., converted to a BED file).

4. Answer

This is the homepage of the link:



Choose a database (Ensembl Genes 113)



Choose dataset:

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Search all species...

New Count Results

URL XML Perl Help

Dataset: Human genes (GRCh38.p14)

Filters: [None selected]

Attributes: Gene stable ID, Gene stable ID version, Transcript stable ID, Transcript stable ID version

Dataset: [None Selected]

In order to maintain service for all users, BioMart browser sessions running for more than 5 minutes are terminated. If you have queries that you think will run longer than this, please choose have the results emailed to you.

29°C Light rain

Search

MSA

9:46 PM 2/4/2025

Filters, choose a region, as described::

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Search all species...

New Count Results

URL XML Perl Help

Dataset: Human genes (GRCh38.p14)

Filters: Chromosome/scaffold: 11, Start: 5240000, End: 5300000

Attributes: Gene stable ID, Gene stable ID version, Transcript stable ID, Transcript stable ID version

Dataset: [None Selected]

REGION:

☒ Chromosome/scaffold

☒ Coordinates

Start: 5240000

End: 5300000

In order to maintain service for all users, BioMart browser sessions running for more than 5 minutes are terminated. If you have queries that you think will run longer than this, please choose have the results emailed to you.

29°C Light rain

Search

MSA

9:49 PM 2/4/2025

Select attributes, using check boxes:

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Search all species...

New Count Results

URL XML Perl Help

Dataset: Human genes (GRCh38.p14)

Filters: Chromosome/scaffold: 11, Start: 5240000, End: 5300000

Attributes: Gene stable ID, Gene stable ID version, Transcript stable ID, Transcript stable ID version, Gene % GC content, CCDS ID, HGNC symbol, HGNC ID

Dataset: [None Selected]

GO

☒ GO term accession

☒ GO term name

☒ GO term definition

GOSlim GOA

☒ GOSlim GOA Accession(s)

External References (max 3)

☒ BioGRID Interaction data, The General Repository for Interaction Datasets ID

☒ CCDS ID

☒ ChEMBL ID

☒ DataBase of Aberrant 3' Splice Sites ID

☒ DataBase of Aberrant 3' Splice Sites name

☒ DataBase of Aberrant 5' Splice Sites ID

☒ DataBase of Aberrant 5' Splice Sites name

☒ EntrezGene transcript name ID

☒ European Nucleotide Archive ID

☒ Expression Atlas ID

☒ GeneCards ID

☒ HGNC symbol

☒ HGNC ID

☒ Human Protein Atlas accession

☒ Human Protein Atlas ID

☐ GO term evidence code

☐ GO domain

☐ GOSlim GOA Description

☐ NCBI gene (formerly Entrezgene) ID

☐ NCBI gene (formerly Entrezgene) accession

☐ PDB ID

☐ Reactome ID

☐ Reactome gene ID

☐ Reactome transcript ID

☐ RefSeq mRNA ID

☐ RefSeq mRNA predicted ID

☐ RefSeq ncRNA ID

☐ RefSeq ncRNA predicted ID

☐ RefSeq peptide ID

☐ RefSeq peptide predicted ID

☐ RFAM ID

☐ RFAM transcript name ID

☐ RNAcentral ID

In order to maintain service for all users, BioMart browser sessions running for more than 5 minutes are terminated. If you have queries that you think will run longer than this, please choose have the results emailed to you.

29°C Light rain

Search

MSA

9:51 PM 2/4/2025

Click “results” (button at upper left), and its shows the results come out with 86402 genes.

The screenshot displays the Ensembl BioMart interface. At the top, the navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. The 'Results' button is highlighted in the top left corner. The main panel shows the 'Dataset 12 / 86402 Genes' and 'Human genes (GRCh38.p14)'. The left sidebar contains filters and attributes. The main panel has a section for 'GENE' with a list of attributes and checkboxes. The bottom status bar shows the system time as 9:53 PM on 2/4/2025.

5. BioMart: working with lists. The goal of this exercise is to access information in BioMart by uploading a text file listing gene identifier of interest.

Follow the steps from problem 2-4, but for the filter set choose Gene (instead of Region), select ID list limit and adjust the pulldown menu to HGNC symbol, then browse for a text file having a list of gene symbols.

See Web Document 2.5 ([WebDocument 2-5 13 humanGlobins HGNCsymbols.txt](#)) for a text file listing official HGNC symbols for 13 human globin genes (*CYGB*, *HBA1*, *HBA2*, *HBB*, *HBD*, *HBE1*, *HBG1*, *HBG2*, *HBM*, *HBQ1*, *HBZ*, *MB*, *NGB*).

You could also enter these gene symbols manually.

For attributes choose any set of features that is different than in problem 2-4, so that you can further explore BioMart resources.

5. Answer

The screenshot shows the Ensembl BioMart interface. The 'Dataset' is 'Human genes (GRCh38.p14)'. The 'Filters' section is expanded, showing 'Gene stable ID(s)' with a list of 13 human globin genes (CYGB, HBA1, HBA2, HBB, HBD, HBE1, HBG1, HBG2, HBM, HBQ1, HBZ, MB, NGB) entered. The 'Attributes' section is also expanded, showing 'Gene % GC content' and 'Gene stable ID'. The 'Results' section shows 'None Selected'. The interface includes a search bar, navigation tabs, and a footer with system information.

Results: (86402genes match)

[BLAST/BLAT](#) | [VEP](#) | [Tools](#) | [BioMart](#) | [Downloads](#) | [Help & Docs](#) | [Blog](#)

[Login/Register](#)

[New](#) | [Count](#) | [Results](#)

[URL](#) | [XML](#) | [Perl](#) | [Help](#)

Dataset 86402 / 86402 Genes
Human genes (GRCh38.p14)

[Filters](#)
Gene stable ID(s) [e.g. ENSG00000000003]: [ID-list specified]

[Attributes](#)
Gene % GC content
Gene stable ID

[Dataset](#)
[None Selected]

Choose File No file chosen

GENE:

☐ Limit to genes (external references)...

☒ Input external references ID list [Max 500 advised]

With HGNC Symbol ID(s)

Gene stable ID(s) [e.g. ENSG00000000003]

☐ Only
☐ Excluded

Choose File textfile.txt

☐ Limit to genes (microarray probes/probesets)...

☐ Input microarray probes/probesets ID list [Max 500 advised]

With AFFY HC G110 probe ID(s)

AFFY HC G110 probe ID(s) [e.g. 737_at]

☐ Only
☐ Excluded

Choose File No file chosen

☐ Transcript count...

In order to maintain service for all users, BioMart browser sessions running for more than 5 minutes are terminated. If you have a session that you think will run longer than this, please choose how the session should be handled.

28°C
Rain showers

Search

MSA

10:03 PM
2/4/2025

14

6. Accessing information from Ensembl.

- (1) Visit the Ensembl resource for humans (via <http://www.ensembl.org/human>).
- (2) In the main search box enter 11:5,240,001-5,300,000. The resulting page displays several panels. At the top, all of chromosome 11 is shown. Where on the chromosome is the region we have selected? In what chromosomal band does this region reside?
- (3) The next panel shows the region in detail. What is the size of the displayed region, in base pairs? In general, genes encoding olfactory receptors are named OR followed by a string of numbers and letters (e.g., *OR51F1*). Approximately how many olfactory receptor genes flank the 60 kb region we have selected? Can you determine exactly how many ORs are in that region?
- (4) Next, we see the region we selected (11:5240001-5300000). Note that there are horizontal tracks (similar to the UCSC Genome Browser).

6. Answer

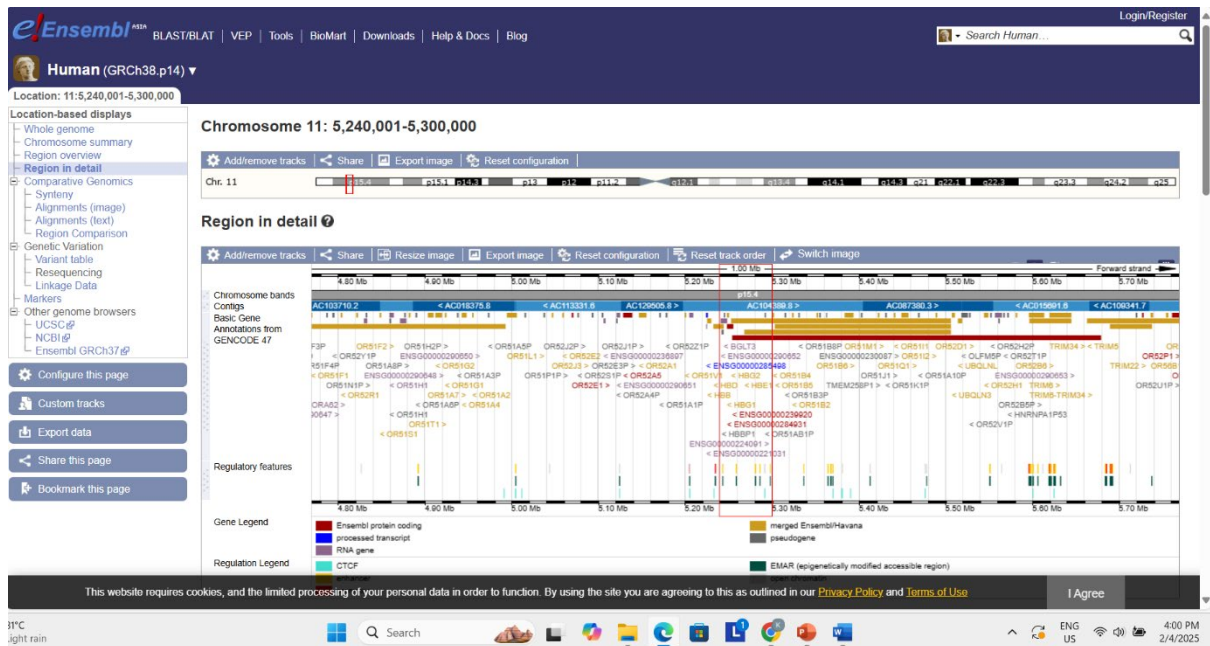
At the search box, enter 11:5,240,001-5,300,000 at human categories.

The screenshot displays the Ensembl genome browser interface for the human genome (GRCh38.p14). The search bar at the top contains the coordinates 11:5,240,001-5,300,000. The main content area is divided into several panels:

- Search Human (Homo sapiens):** Shows the search bar with the coordinates 11:5,240,001-5,300,000 and a 'Go' button. Below it, there is a link to 'e.g. BRCA2 or 17:63992802-64038237 or rs699 or osteoarthritis'.
- Genome assembly: GRCh38.p14 (GCA_000001405.29):** Includes links for 'More information and statistics', 'Download DNA sequence (FASTA)', 'Convert your data to GRCh38 coordinates', and 'Display your data in Ensembl'. It also shows 'Other assemblies' with a dropdown menu set to 'GRCh37 Full Feb 2014 archive with BLAST, VEP and BioMart'.
- Gene annotation:** Describes what can be found (protein-coding and non-coding genes, splice variants, cDNA and protein sequences, non-coding RNAs) and provides links for 'More about this genebuild', 'Download FASTA files for genes, cDNAs, ncRNA, proteins', 'Download GTF or GFF3 files for genes, cDNAs, ncRNA, proteins', and 'Update your old Ensembl IDs'. It includes an 'Example gene' (Pax6) and an 'Example transcript' (FOXP2).
- Comparative genomics:** Describes what can be found (homologues, gene trees, and whole genome alignments across multiple species) and provides links for 'More about comparative analysis' and 'Download alignments (EMF)'. It includes an 'Example gene tree'.
- Variation:** Describes what can be found (short sequence variants and longer structural variants, disease and other phenotypes) and provides links for 'More about variation in Ensembl', 'Download all variants (GVF)', and 'Variant Effect Predictor (VEP)'. It includes an 'Example variant' (ATCGAGCT/ATCAGCT/ATCGAGT) and an 'Example' (eye).
- Regulation:** Describes what can be found (regulatory features like enhancers and promoters, and regulatory activity) and includes an 'Example' (eye).

The bottom of the page features a footer with a cookie notice, a language selector (ENG US), and a timestamp (3:59 PM 2/4/2025).

Here is the upper portion of the results page:



Here shows many olfactory receptor (OR) genes.

7. Accessing information from UCSC

Hemoglobin is a tetramer composed primarily of two alpha globin subunits and two beta globin subunits.

Consider alpha globin.

There are two related human genes (official gene symbols HBA1 and HBA2).

- a) Use the UCSC Genome Browser (<http://genome.ucsc.edu/>) to determine the length of the intergenic region between the HBA1 and HBA2 genes.

7. Answer

Enter HBA1 at the search bar:

The screenshot shows the UCSC Genome Browser homepage. The search bar at the top contains the text "HBA1". Below the search bar, there are two main sections: "Tools" and "News". The "Tools" section lists several tools: Genome Browser, BLAT, In-Silico PCR, Table Browser, LiftOver, REST API, and Variant Annotation Integrator. The "News" section lists several updates: Mar. 31, 2025 - New Pseudogenes track for hg38; Mar. 14, 2025 - denovo-db tracks for hg19 now available; Mar. 06, 2025 - New MITOMAP track for hg38 and hg19; Mar. 05, 2025 - New Splicing Impact super track and SpliceVarDB track for hg38; Mar. 03, 2025 - New highlight features for track hubs now available; Feb. 24, 2025 - enGenome VarChat track for human (hg38 and hg19). At the bottom, there are announcements for meetings and workshops: Biocuration 2025 -- Kansas City, Missouri, April 7-9, 2025; CSHL Biology of Genomes -- Cold Spring Harbor, NY, May 6-10, 2025; and ESHG: European Human Genetics -- Milan, Italy, May 24-27, 2025. The page also features a "Meetings and Workshops: Come see us in person!" banner.

It will show:

Search across the Genome Browser site

Search:

Human Feb. 2009 (GRCh37/hg19)

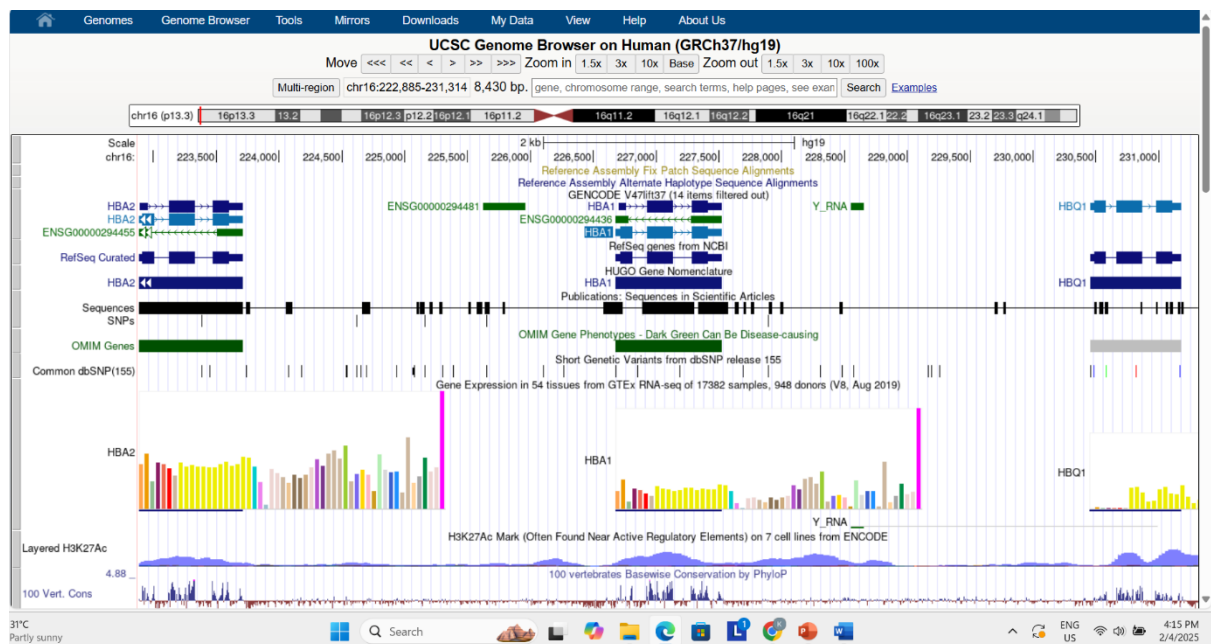
Use the tree to hide/show results from only these categories. Hover your mouse over each category for an explanation:

- hg19 Track Data (187 results)
 - Visible Tracks (4 results)
 - HGNC (1 results)
 - RefSeq Curated (1 results)
 - GENCODE V47lift37 (2 results)
 - Currently Hidden Tracks (183 results)
 - Genes and Gene Predictions (89 results)
 - Other RefSeq (8 results)
 - Ensembl Genes (4 results)
 - NCBI RefSeq (1 results)
 - GENCODE Versions (4 results)
 - Vega Genes (4 results)
 - Phenotypes, Variants, and Literature (1 results)
 - GeneReviews (1 results)
 - Variation (3 results)
 - gnomAD (2 results)
 - mRNA and EST (90 results)
 - Human mRNAs (10 results)
 - Other mRNAs (80 results)

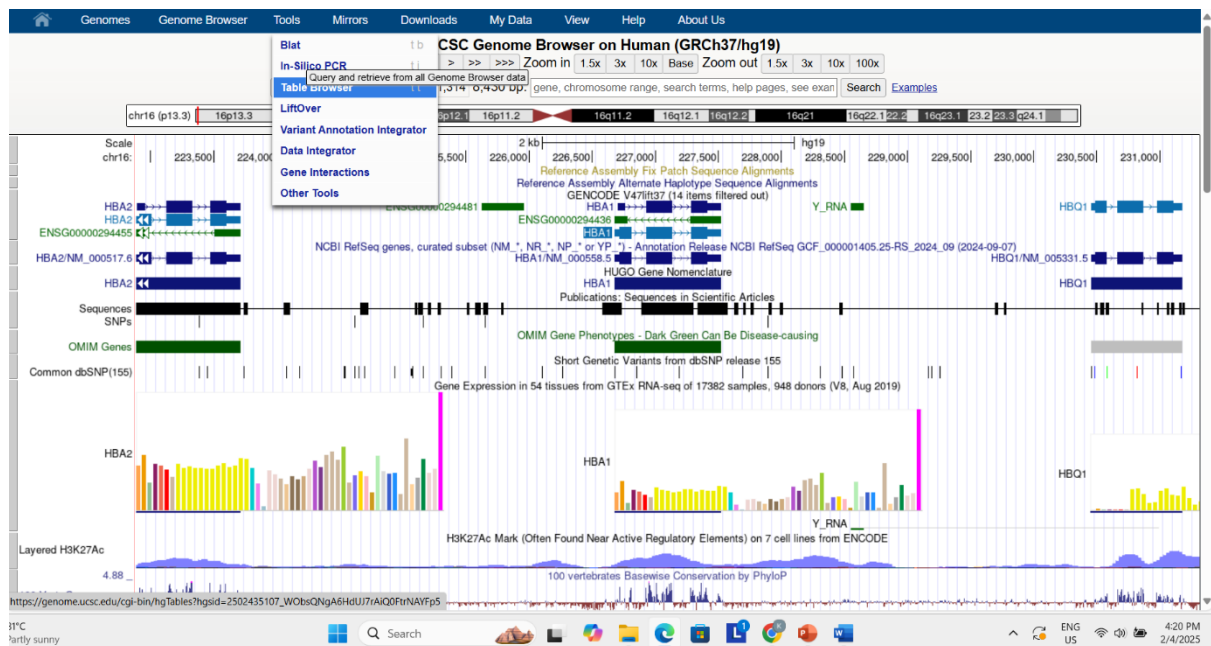
Search Results on hg19 (Human Feb. 2009 (GRCh37/hg19))

- ☐ HUGO Gene Nomenclature:
 - HBA1** - chr16:226680-227521
- ☐ Gencode Genes:
 - HBA1** (ENST00000320868.9_4) - chr16:226679-227521 - hemoglobin subunit alpha 1 (from RefSeq NM_000558.5)
 - HBA2** (ENST00000251595.11_5) - chr16:222875-223709 - hemoglobin subunit alpha 2 (from RefSeq NM_000517.6)
- ☐ NCBI RefSeq genes, curated subset (NM_*, NR_*, NP_* or YP_*):
 - NM_000558.5** - chr16:226679-227521
- ☐ RefSeq Genes:
 - HBA1** - chr16:226679-227521 - (NM_000558) hemoglobin subunit alpha
- ☐ Non-Human RefSeq Genes:
 - HBA1** - chr16:222875-223710 - (NM_001132429) hemoglobin subunit alpha
 - HBA1** - chr16:226693-227508 - (NM_001266776) hemoglobin subunit alpha
 - HBA1** - chr16:222889-223698 - (NM_001266776) hemoglobin subunit alpha
 - HBA1** - chr16:226713-227416 - (NM_001004376) hemoglobin subunit alpha-A
 - hba1** - chr16:226926-227410 - (NM_203529) hemoglobin subunit alpha
 - HBA1** - chr16:226679-227520 - (NM_001042626) hemoglobin subunit alpha
 - HBA1** - chr16:226693-227520 - (NM_001168816) hemoglobin subunit alpha
 - hba1.S** - chr16:226926-227410 - (NM_001088024) hemoglobin subunit alpha-2
- ☐ Basic Gene Annotation Set from GENCODE Version 47lift37 (Ensembl 113):
 - HBA1** - chr16:226679-227521
 - HBA1** - chr16:226703-227521
- ☐ Comprehensive Gene Annotation Set from GENCODE Version 47lift37 (Ensembl 113):
 - HBA1** - chr16:226679-227521
 - HBA1** - chr16:226697-227519
 - HBA1** - chr16:226703-227521
 - HBA1** - chr16:226747-227273
- ☐ Basic Gene Annotation Set from GENCODE Version 46lift37 (Ensembl 112):
 - HBA1** - chr16:226679-227521
 - HBA1** - chr16:226703-227521
- ☐ Comprehensive Gene Annotation Set from GENCODE Version 46lift37 (Ensembl 112):
 - HBA1** - chr16:226679-227521

Click on HBA1 and zoom out 10-fold and you will see the adjacent HBA1 and HBA2 genes:



To get a more precise answer use the Table Browser, from the Tool menu:



Set a region encompassing both genes and set the various menus as shown here:

The screenshot shows the Table Browser configuration page. The settings are as follows:

- Select dataset:**
 - Clade: Mammal
 - Genome: Human
 - Assembly: Feb. 2009 (GRCh37/hg19)
 - Group: Genes and Gene Predictions
 - Track: NCBI RefSeq
 - Table: RefSeq All (ncbiRefSeq)
- Define region of interest:**
 - Region: ☐ Genome ☐ ENCODE Pilot regions
 - Position: chr16:222,885-231,314
- Optional: Subset, combine, compare with another track:**
 - Filter: Create
 - Subtrack merge: Create
 - Intersection: Create
 - Correlation: Create
- Retrieve and display data:**
 - Output format: All fields from selected table
 - Send output to: ☐ Galaxy ☐ GREAT
 - Output filename: (add .csv extension if opening in Excel, leave blank to keep output in browser)
 - Output field separator: ☒ tsv (tab-separated) ☐ csv (for excel)
 - File type returned: ☒ Plain text ☐ Gzip compressed

This is the output (set to all fields):

#bin	name	chrom	strand	txStart	txEnd	cdsStart	cdsEnd	exonCount	exonStarts	exonEnds	score	name2	cdsStartStat	cdsEndStat	exonFrames
586	NM_000517.6	chr16	+	222874	223709	222911	223599	3	222874,223123,223470,	223006,223328,223709,	0	HBA2	cmpl	cmpl	0,2,0,
586	NM_000558.5	chr16	+	226678	227521	226715	227410	3	226678,226927,227281,	226810,227132,227521,	0	HBA1	cmpl	cmpl	0,2,0,
586	NM_005331.5	chr16	+	230457	231178	230485	231107	3	230457,230664,230978,	230580,230869,231178,	0	HBQ1	cmpl	cmpl	0,2,0,

This is the output (as a BED file). From either of these outputs you can calculate the exact intergenic distance.

chr16	222874	223709	NM_000517.6	0	+	222911	223599	0	3	132,205,239,	0,249,596,
chr16	226678	227521	NM_000558.5	0	+	226715	227410	0	3	132,205,240,	0,249,603,
chr16	230457	231178	NM_005331.5	0	+	230485	231107	0	3	123,205,200,	0,207,521,

Take the beginning of one the more 3' gene (position 226,715) and subtract the end of the more 5' gene (position 223,599) = 3,116 base pairs.

8. Accessing information from UCSC. The purpose of this exercise is for you to gain familiarity with the UCSC Genome Browser

a) What types of repetitive DNA elements occur in the human beta globin gene?

The purpose of this exercise is for you to gain familiarity with the UCSC Genome Browser.

As a user, you choose which tracks to display. Visit and explore as many as possible. Try to get a sense for the main categories of information offered at the Genome Browser.

As you work in the genome browser you may want to switch between builds hg18, hg19, and hg20. To do so, go the “View” pull-down menu and use “In other genomes (convert).”

Follow the following steps.

(1) Go to <http://genome.ucsc.edu/cgi-bin/hgGateway> . Make sure the clade is Mammal, genome is Human, assembly is NCBI36/hg19, and in the “gene” box enter hbb for beta globin. Click Submit. Note that HBB is the official gene symbol for beta globin, but you can use the lowercase hbb for this search. Use NCBI Gene (or <http://www.genenames.org> for the HGNC site) to find the official gene symbol of your favorite gene.

(2) Click the “default tracks” button. Note the position you have reached (chromosome 11, spanning 1,606 base pairs close to the beginning of the short or “p” arm of the chromosome). Note the appearance of over a dozen graphical tracks that are horizontally oriented.

(3) One of the tracks is “Repeating Elements by Repeatmasker.” There are two black blocks. Right click on the block and select “Full.” Alternatively, scroll down to the section entitled “Variation and Repeats,” locate “RepeatMasker,” and change the pull-down menu setting from “dense” to “full.” Note also that by clicking the blue heading “RepeatMasker” you visit a page describing the RepeatMasker program and its use at the UCSC Genome Browser.

(4) View the RepeatMasker output.

Choose one answer.

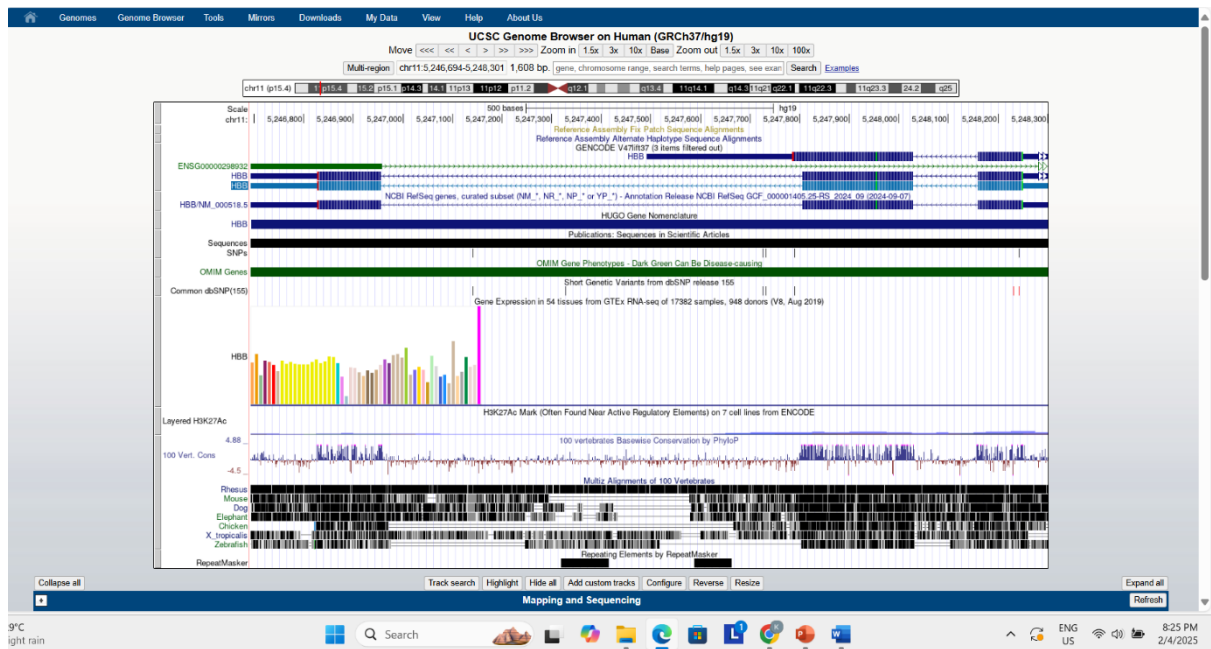
- There are no repetitive elements.
- There is one SINE element and one LINE element.
- There is one LTR and one satellite.
- There is one LINE element and one low complexity element.
- There are well over a dozen repetitive elements.

8. Answer:

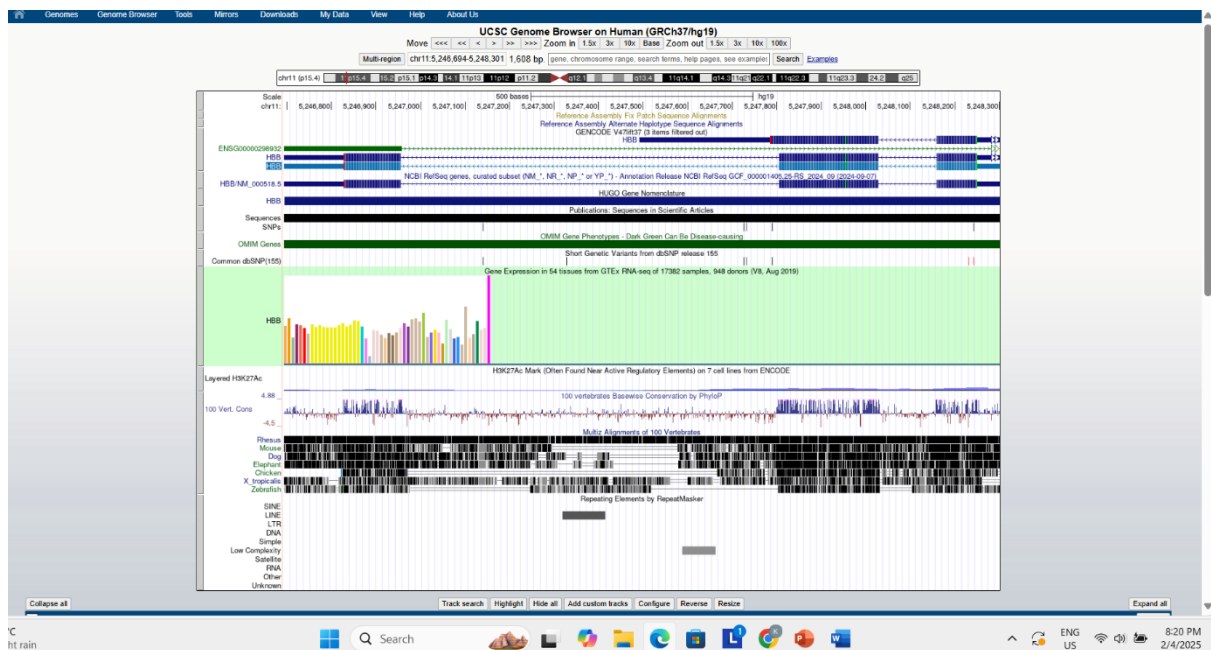
Enter HBB in search bar:

The screenshot displays the UCSC Genome Browser Gateway interface. At the top, the 'Find Position' section shows 'Human Assembly' and 'Feb. 2009 (GRCh37/hg19)' selected. The 'Position/Search Term' field contains 'HBB'. Below this, the 'Human Genome Browser - hg19 assembly' section provides information about the February 2009 human reference sequence. A 'Sample position queries' section lists various queries and their corresponding genome browser responses. The 'Assembly Details' section at the bottom provides additional information about the assembly. The main track area shows the HBB gene structure with various annotations, including RepeatMasker tracks showing repetitive elements.

The view of default tracks:



The view with the RepeatMasker track(at bottom) expanded:



By inspection answer (d) is correct.

9. Accessing information from the UCSC Table Browser.

a) How many SNPs span the human beta globin gene?

To solve this problem, use the UCSC Table Browser. The Table Browser is equally useful as the Genome Browser. Instead of offering visual output, it offers tabular output. Often it is not practical (or accurate) to visually count elements from the Genome Browser. We often want quantitative information about genomic features in some chromosomal region or across the whole genome. This problem asks about single nucleotide polymorphisms (SNPs), which are positions that vary (i.e., exhibit polymorphism) across individuals in a population. Follow the following steps.

- (1) Start at the HBB region of the UCSC Genome Browser and click the “Tables” tab along the top. Alternatively, you can go to the UCSC website (<http://genome.ucsc.edu>), and click Tables. Set the clade to Mammal, genome to Human, assembly to NCBI36/hg18, group to Variation and Repeats, track to SNPs(130), table to snp130, and region to position chr11:5203272-5204877. Note that if the position is not already set, you can type hbb into the position box, click “lookup” and the correct position will be entered.
- (2) To see the answer to this problem, click “summary/statistics.” The item count tells you how many SNPs there are.
- (3) To see the answer as a table, set the output format to “all fields from selected table,” make sure the “Send output to Galaxy / GREAT” boxes are not checked, and click the “get output” box. The SNPs are shown as a table including chromosome, start, and stop position.
- (4) Try the various output options, such as a bed file or a custom track. Note that you can output the information as a file saved to your computer.

9. Answer:

In the group “Variation and repeat” the tracks that are currently available include SNPs(130):

Table Browser

Use this tool to retrieve and export data from the Genome Browser annotation track database. You can limit retrieval based on data attributes and intersect or merge with data from another track, or retrieve DNA sequence covered by a track. [More...](#)

Select dataset

Clade: Mammal Genome: Human Assembly: Mar. 2006 (NCBI36/hg18)
 Group: Variation and Repeats Track: SNPs (130)
 Table: snp130 [Data format description](#)

Note: Most dbSNP tables are huge. Trying to download them through the Table Browser usually leads to a timeout. Please see our [Data Access FAQ](#) on how to download dbSNP data.

Define region of interest

Region: ☐ genome (unavailable for selected track) ☐ ENCODE Pilot regions ☒ Position chr11:5,203,272-5,204,877 [Lookup](#) [Define regions](#)

Note: This track is unavailable for genome-wide download. Usually, this is due to distribution restrictions of the source database or the size of the track. Please see the track documentation for more details. Contact us if you are still unable to access the data.

Identifiers (names/accessions): [Paste list](#) [Upload list](#)

Optional: Subset, combine, compare with another track

Filter: [Create](#)
 Intersection: [Create](#)
 Correlation: [Create](#)

Retrieve and display data

Output format: BED - browser extensible data Send output to ☒ [Galaxy](#) ☐ [GREAT](#)
 Output filename: (leave blank to keep output in browser)
 File type returned: ☒ Plain text ☐ Gzip compressed

[Get output](#) [Summary/statistics](#)

For SNPs(130), clicking summary/statistics shows :

SNPs (130) (snp130) Summary Statistics

item count	502
item bases	1,606 (100.00%)
item total	23,807 (1482.38%)
smallest item	0
average item	47
biggest item	92,848

Region and Timing Statistics

region	chr11:5,203,272-5,204,877
bases in region	1,606
bases in gaps	0
load time	0.01
calculation time	0.00
free memory time	0.00
filter	off
intersection	off

10. Accessing information from Galaxy

a) How big is the largest RefSeq gene on human chromosome 21? Solve this problem by using Galaxy.

- (1) First go to Galaxy (<http://usegalaxy.org>). Optionally, you can register (under the “User” tab).
- (2) On the left sidebar, choose “Get Data” then “UCSC Main table browser.”
- (3) Set the clade (Mammal), genome (Human), assembly (GRCh37 [or try GRCh38]), group (Genes and Gene Prediction Tracks), track (RefSeq Genes), table (RefGene), region (click position then enter “chr21” without the quotation marks) then click “lookup” right next to the position. Under output format choose “BED-browser extensible data” and click the box “Send output to Galaxy.”
- (4) Optionally, click “summary/statistics” to get a quick look at how many proteins are assigned to chromosome 21. (That answer is currently 636.)
- (5) At the lower left part of the page, click “get output.” Note that you now have a variety of output options; choose BED and click “Send query to Galaxy.”
- (6) Galaxy’s central panel informs you that the job is added to the queue.
- (7) Your data set is available in the history panel to the right. Click the data set header [1: UCSC Main on Human: refGene (chr21:1-46944323)] to see the number of regions and to see the column headers. Click the “eye” icon to see your data in the central panel.
- (8) Next figure out the size of the genes. First, add a new column. On the left Galaxy panel click “Text Manipulation” then “Compute an expression on every row.” Add the expression $c3 - c2$ to take the end position of each gene and subtract the beginning. For “Round result?” choose “Yes.” Click “Execute.”
- (9) A new data set is created, called “Compute on data 1.” There is a new column 13 with the sizes of all the genes. Go to the left sidebar of Galaxy, click “Filter and Sort,” click “Sort data in ascending or descending order” and choose the query; the column (c13); the flavor (numerical sort); the order (descending); and click Execute.
- (10) A new data set is created. Click the eye icon to see your spreadsheet in the main Galaxy panel. Your answer is there on the first (top) row. Optionally, go to “Text Manipulation,” select “Cut columns from a table,” and Cut columns (c5,c6,c7,c8,c9,c10,c11,c12). This will clean up your table, making it easier to see column 13 with the gene lengths.

10. Answer:

Here is the Galaxy home page:

The screenshot shows the Galaxy web interface. On the left is a sidebar with navigation links: Upload, Tools, Workflows, Visualization, Histories, and Pages. The 'Tools' section is expanded, showing a list of tools including GDCWebApp, IEDB MHC Binding prediction, InterMine server, NCBI Accession Download, NCBI Datasets Gene, NCBI Datasets Genomes, Protein Database Downloader, SRA server, UCSC Archaea table browser, UCSC Main table browser, UniProt, and Upload File. The main content area features a welcome message, a GBCC2025 abstract submission announcement, and a 'Donate to the James P. Taylor Foundation for Open Science' button. On the right is a 'History' panel showing an empty history list with a message: 'This history is empty. You can load your own data or get data from an external source.'

Set the Table Browser entries as follows:

The screenshot shows the Galaxy Table Browser interface. The top navigation bar includes links for Genomes, Genome Browser, Tools, Mirrors, Downloads, My Data, Projects, Help, and About Us. The 'Table Browser' section is active, displaying instructions and configuration options. The 'Select dataset' section is set to 'Mammal' for Clade, 'Human' for Genome, 'Feb. 2009 (GRCh37/hg19)' for Assembly, 'Genes and Gene Predictions' for Group, 'NCBI RefSeq' for Track, and 'RefSeq All (ncbiRefSeq)' for Table. The 'Define region of interest' section is set to 'Region: chr21:1-48,129,895'. The 'Optional: Subset, combine, compare with another track' section has 'Filter', 'Subtrack merge', 'Intersection', and 'Correlation' all set to 'Create'. The 'Retrieve and display data' section is set to 'Output format: BED - browser extensible data', 'Send output to: Galaxy', and 'File type returned: Plain text'. The 'Using the Table Browser' section is also visible at the bottom.

The current total number of genes (989) on this chromosome is shown by clicking the summary/statistics tab:

Genomes Genome Browser Tools Mirrors Downloads My Data Projects Help About Us

ncbiRefSeq (ncbiRefSeq) Summary Statistics

item count	989
item bases	15,752,830 (44.87%)
item total	73,954,284 (210.64%)
smallest item	49
average item	74,777
biggest item	836,159
block count	9,288
block bases	1,063,389 (3.03%)
block total	3,073,880 (8.76%)
smallest block	9
average block	331
biggest block	18,112

Region and Timing Statistics

region	chr21:1-48,129,895
bases in region	48,129,895
bases in gaps	13,021,193
load time	0.01
calculation time	0.01
free memory time	0.00
filter	off
intersection	off

29°C Rain showers 8:38 PM 2/4/2025

Click “get output” then send query to Galaxy.

Genomes Genome Browser Tools Mirrors Downloads My Data Projects Help About Us

Output ncbiRefSeq as BED

☐ Include custom track header:

name=

description=

visibility=

url=

Create one BED record per:

☒ Whole Gene

☐ Upstream by bases

☐ Exons plus bases at each end

☐ Introns plus bases at each end

☐ 5' UTR Exons

☐ Coding Exons

☐ 3' UTR Exons

☐ Downstream by bases

Note: if a feature is close to the beginning or end of a chromosome and upstream/downstream bases are added, they may be truncated in order to avoid extending past the edge of the chromosome.

9°C Rain showers 8:40 PM 2/4/2025

Follow the above directions. Imported data appear in the right sidebar; click the eye icon to see the data in the main display:

The screenshot shows the Galaxy web interface. On the left, the 'Tools' menu is open, showing 'All Tools' and a search bar. The 'GENERAL TEXT TOOLS' section is selected, and 'Compute on rows' is highlighted. The main panel displays a table of genomic data for chromosome 21. The table has columns: Chrom, Start, End, Name, Score, Strand, ThickStart, ThickEnd, ItemRGB, and BlockCount. The data is filtered for chromosome 21 and shows various gene entries with their coordinates and scores. On the right, the 'History' panel shows a list of datasets, including '1: UCSC Main on Human: ncbi RefSeq (chr21:1-48,129,895)'.

As indicated in the problem's directions (above), use Tools > Text Manipulation > Compute on row > add expression c3-c2 (this will take the end position minus the start position of every gene on chromosome 21).

The screenshot shows the Galaxy web interface with the 'Compute on rows' tool configuration. The 'Input file' is set to '1: UCSC Main on Human: ncbi RefSeq (chr21:1-48,129,895)'. The 'Expressions' section shows '1: Expressions' with 'Add expression' set to 'c3-c2'. The 'Mode of the operation' is set to 'Append'. The 'History' panel on the right shows a list of datasets, including '1: UCSC Main on Human: ncbi RefSeq (chr21:1-48,129,895)'.

Here is the output: a new file is created (data set 4: compute on data 3) with a new column 13 showing the size of every gene:

Now we need to sort the results on column 13 to see which is the largest gene on chromosome 21. Tools

```
> Filter and sort. Execute.
```

The answer is that the largest gene is 836159 base pairs.

Galaxy

Using 456.9 KB

Login or Register

Upload

Tools

Workflows

Visualization

Histories

Pages

All Tools

sort

Show Sections

Sort data in ascending or descending order

Sort collection

SortSam sort SAM/BAM dataset

Sort a row according to their columns

Sort.seqs put sequences in different files in the same order

Sort Column Order by heading

Samtools sort order of storing aligned sequences

bedops sort-bed

bedtools SortBED order the intervals

Pairtools sort Sort a 4dn pairs/pairsam file

chainSort Sort chains

Filter with SortMeRNA of ribosomal RNAs in metatranscriptomic data

MiModD Sort takes a SAM/BAM dataset and generates a coordinate/name-sorted copy

VSearch sorting

VCFsort: Sort VCF dataset by coordinate

	Column 13
8100,681810,697454,835619,	836159
8100,681810,697454,835619,	836159
8100,681810,697454,835619,	836159
	561239
	544924
	544924
	544924
	544924
	540289
	540255
	539663
	539663
	487880
	480935
	440665
	440665
	440665
	440665
	440665

History

search datasets

Unnamed history

644 kB

5: Sort on data 2

Add Tags

989 regions 13 columns

format bed, database hg19

1.Chrom 2.Start 3.End 4 5 6

chr21 41382925 42219084 HPI_001389.5 0 -

chr21 41382925 42219084 HPI_001271534.3 0 -

chr21 41382925 42219084 HPI_073202.3 0 -

chr21 17442807 18004046 NR_136541.1 0 +

chr21 22370726 22915650 HPI_001352595.2 0 +

2: Compute on data 1

1: UCSC Main on Human: ncbi RefSeq (chr21:1-48,129,895)

7508_5G

Internet access

29°C

light rain

Search

MSA

9:32 PM

2/4/2025